Experimental Modal Analysis using Impact hammer

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Experimental Modal Analysis (EMA) using impact hammer.

Perform experimental modal analysis of a frame structure (as shown in figure 1.) using impact hammer to find natural frequency and mode shapes of the frame structure. Here, only steel tip was used as impact hammer tip.

Figure 1: Frame structure

Impact hammer calibration

Impact hammer is a widely used instrument for EMA of a structures. It has its application mainly in aerospace and mechanical components such as turbine blades, aircraft wings etc. For the purposes of this lab work, the steel tip impact hammer provided sufficient and broadband excitation for our frequencies of interest.

The appropriate calibration of impact hammers is critical to ensure measurement accuracy and meet quality management requirements.

The following steps were employed to calibrate the impact hammer:

Step 1: The mass of the calibration system was determined by measuring the mass of suspended object and any supporting components. In our case the mass of supporting wire was neglected.

$$m = m_{\text{object}} + m_{\text{string}} \tag{1}$$

Step 2: An accelerometer was securely placed on the mass, ensuring proper alignment.

Step 3: An impulsive force was applied to the suspended mass using the impact hammer. The acceleration response of the mass and the corresponding voltage signal from the hammer were recorded.

Step 4: Using the recorded acceleration data, peak acceleration response was determined and the corresponding impact force was calculated as follows.

$$F = m\, a_{\text{peak}} \tag{2}$$

Step 5: The peak voltage from hammer signal was obtained and the calibration constant was determined as follows.

$$K = \frac{F}{V} \tag{3}$$

We have the voltage recording and the value of F so by dividing both we get calibration constant of the impact hammer.

This calibration procedure was performed two times and their average was taken to get the final calibration constant.

Table 1: Calibration Results

Calibration 1			Calibration 2		
Parameter	**Value**	**Unit**	**Parameter**	**Value**	**Unit**
Max Acceleration	3.980121512	m/s^2	Max Acceleration	10.18547517	m/s^2
Mass	1.25	kg	Mass	1.25	kg
F (Force)	4.97515189	N	Force	12.73184396	N
V	727.5609759		V	1248.414634	
K	**0.006838124**		K	**0.01019841**	

The final calibration constant was **0.00852**.

The structure was excited floor by floor using the impact hammer, with the input force and corresponding responses recorded at each degree of freedom (DOF). This process was repeated sequentially until excitation and response data were obtained at all floor levels. The applied force at the top floor and its corresponding structural responses are presented in the figure below. *Only one instance of excitation and response is shown here for the cohesiveness of the report.*

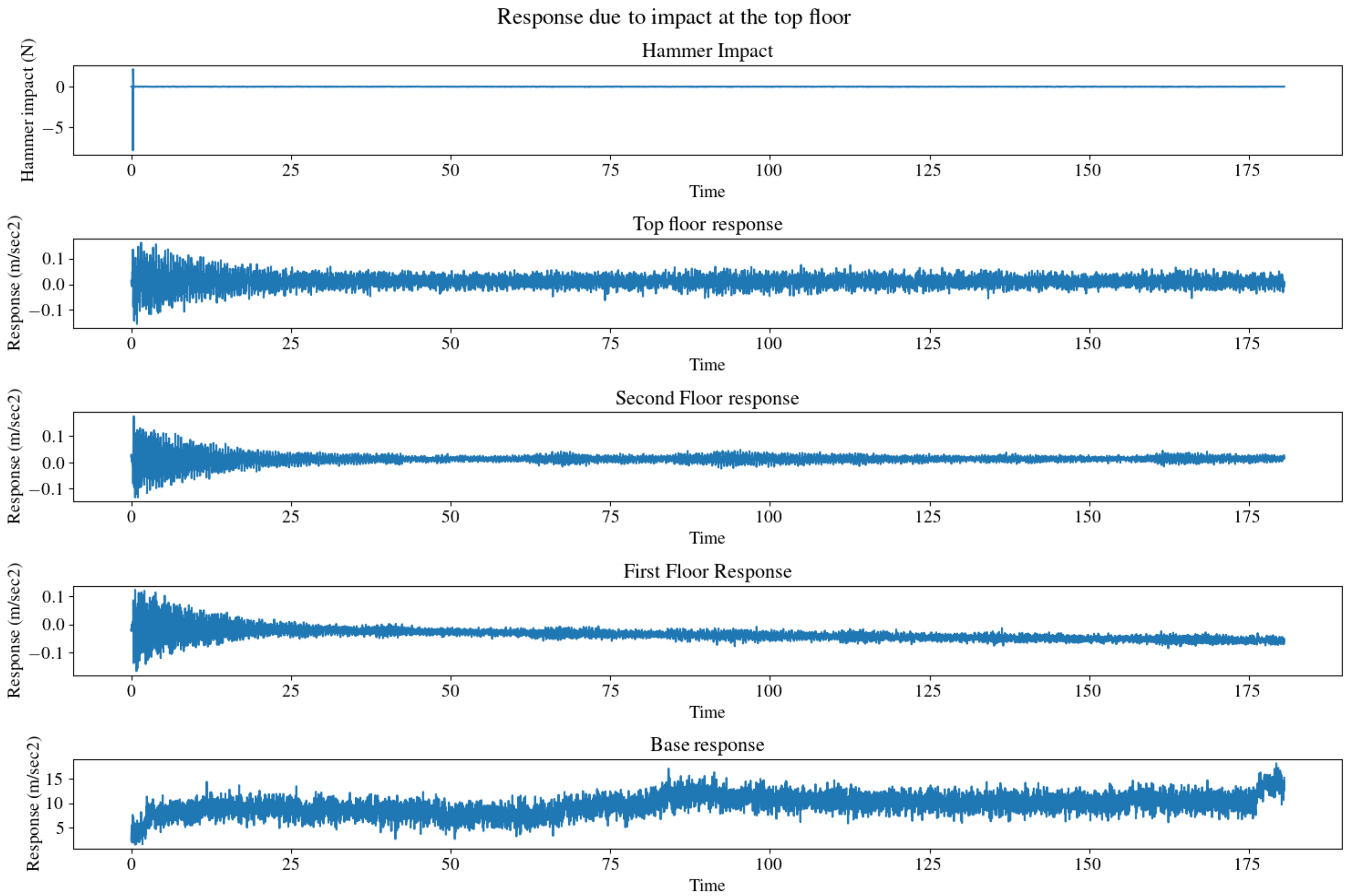

Figure 2: Impact load at top floor and its corresponding response

From Figure 2, we can observe some excitation in the base of the structure. This is likely attributed to the fact that the frame is mounted on a shaker platform, which was not completely stationary during testing. Hence, we observe some movement.

Frequency Response Function and Matrix:

The outputs can be represented in Frequency response matrix as shown in figure below.

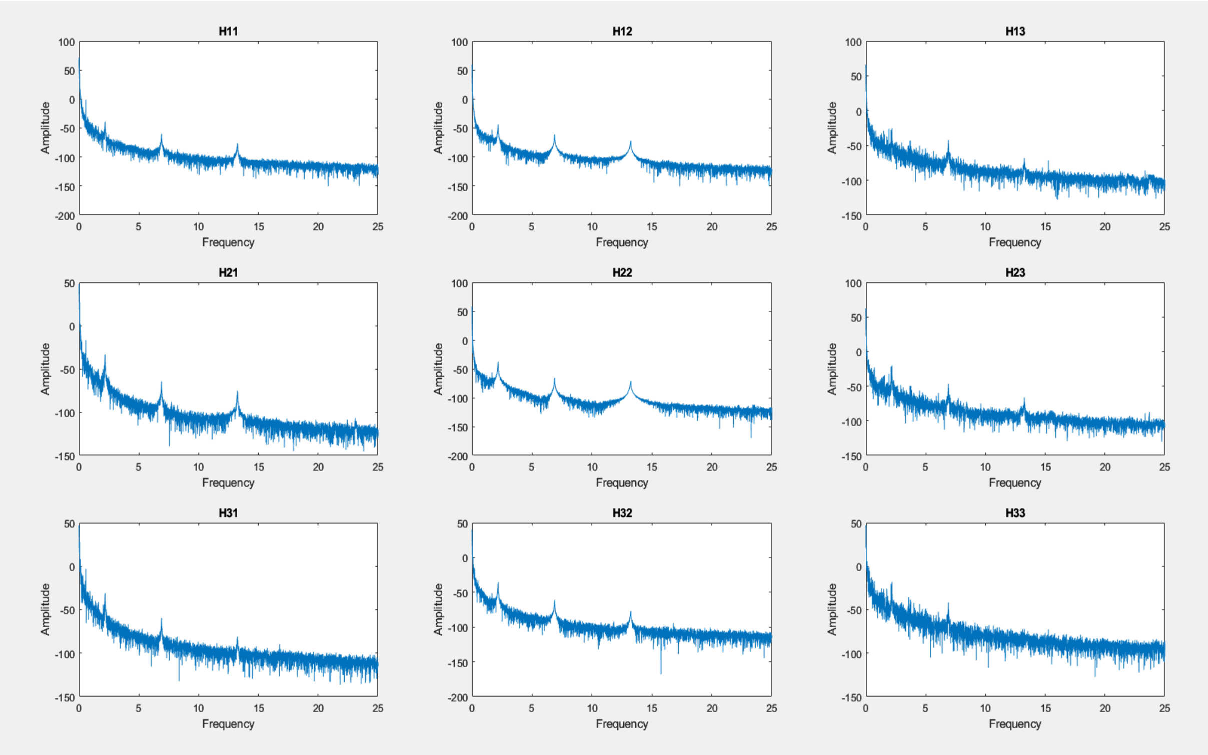

Figure 3: Frequency Response Matrix

From these plots after peak picking we can confirm the frequency of 1st, 2nd and 3rd mode to be:

Table 2: Frequencies of each mode.

Mode 1	Mode 2	Mode 3
2.17 Hz	6.89 Hz	13.263 Hz

One instance of peak picking at H11 is shown below:

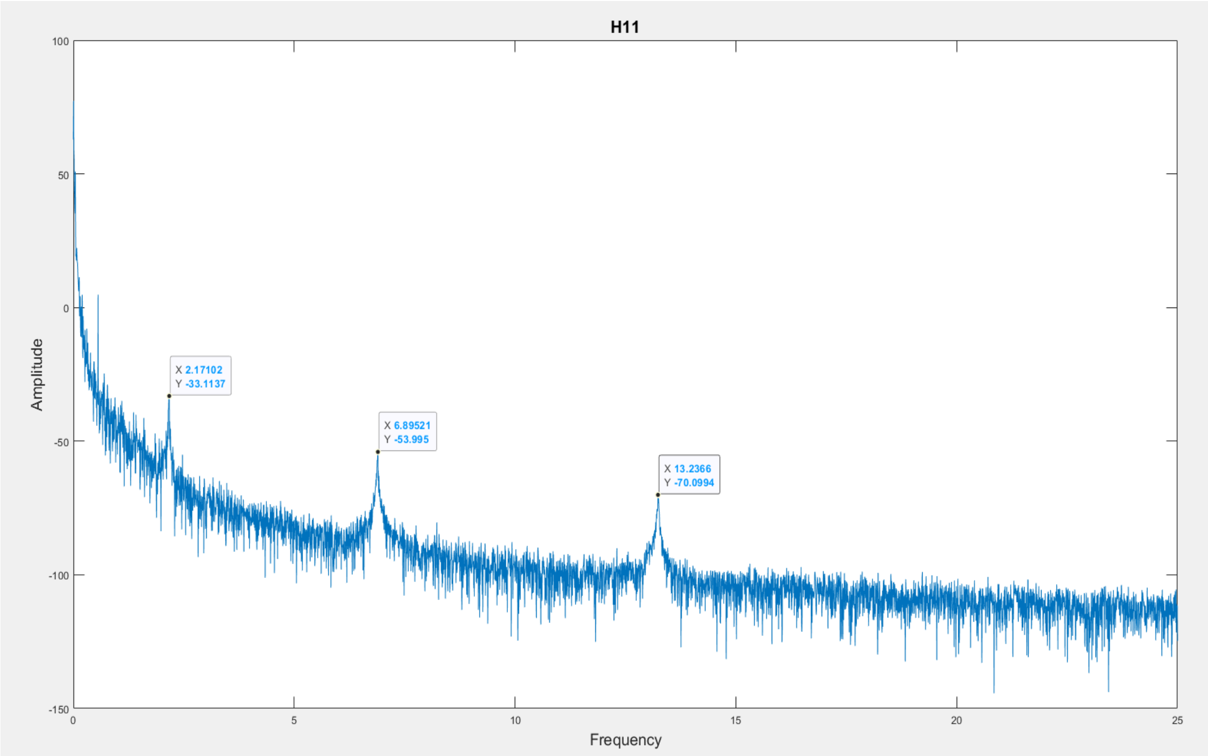

Figure 4: Peak picking the modal frequencies in H11 frf

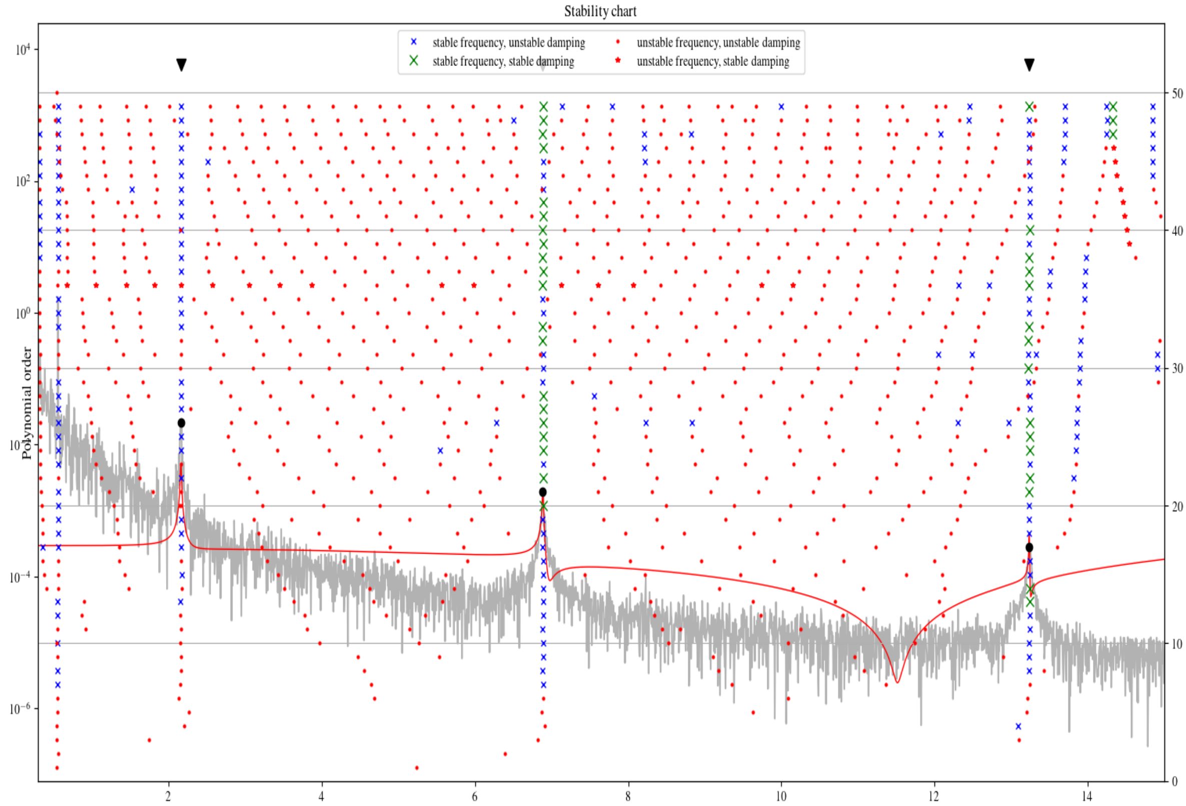

Figure 5: Stabilization plot for H11 frf

The frequencies values were validated with other plots as well but are not presented in the report to keep the report concise and free of unnecessary repetition.

Imaginary portion of FRF

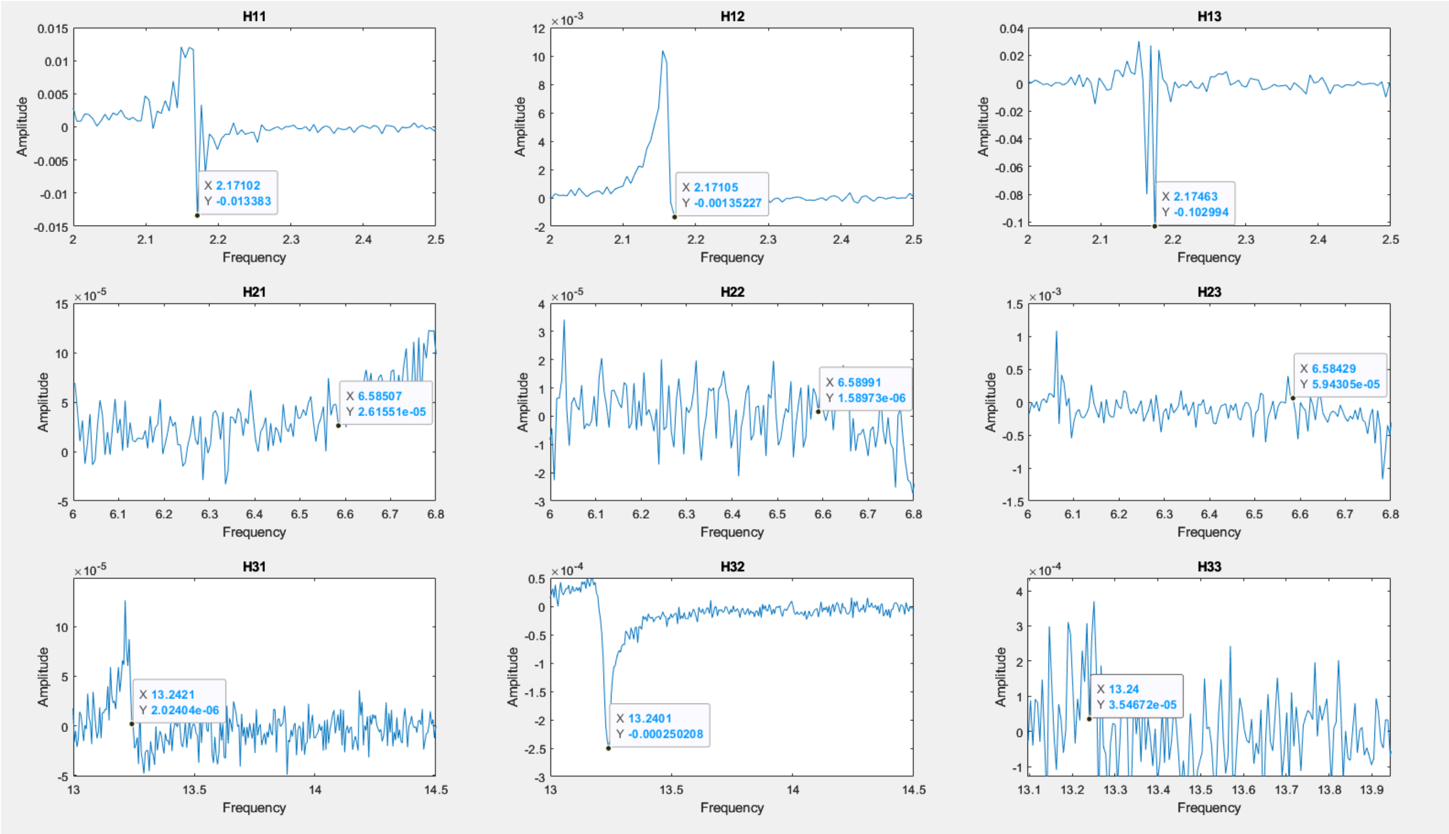

Figure 6: Imaginary Protion of FRF and mode shapes

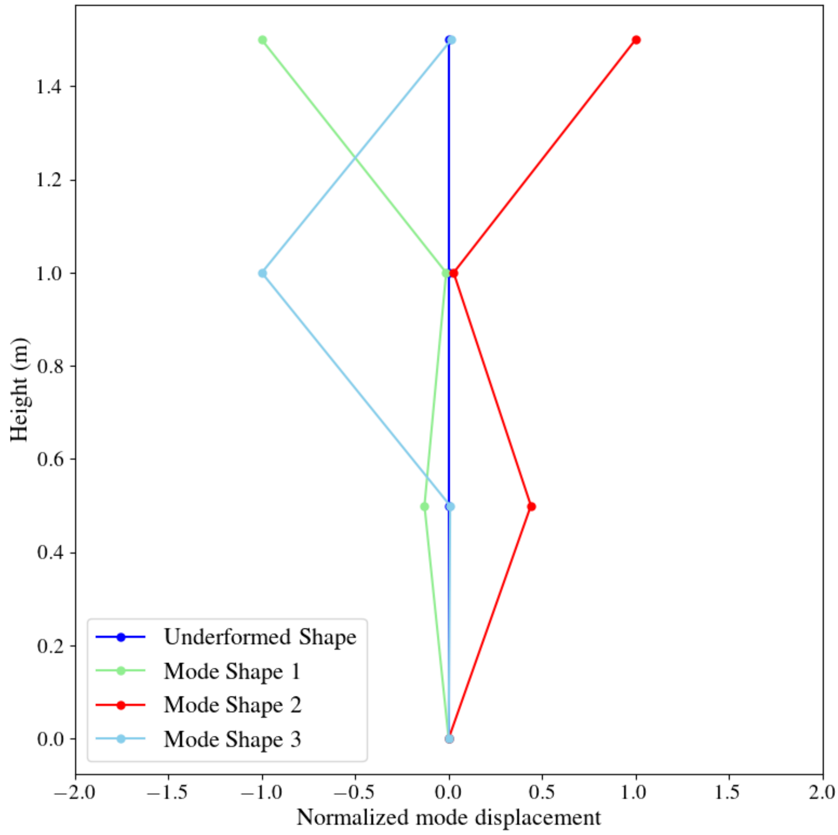

Figure 7: Mode Shape

Comments: Here we observe that the mode shapes are sporadic and do not exhibit expected behavior. This is also observed in the upcoming case of mass added EMA as well. This is suspected to be following reasons:

- **Calibration error**: The impact hammer caibration was performed only twice yeilding a calibration constants of value : 0.00684 and 0.01020. These values are far from each other and Averaging only two such values is insufficient for the mean to converge toward the true calibration constant. This erroneous scaling of the measured force signal directly corrupts the FRF amplitudes, and consequently, the mode shapes extracted from the imaginary portion of the FRF.

- **Sensor error**: From the time history plot it can also be observed that one of the sensors is malfunctioning and recording amplitude at a higher degree than others. So, it can be suspected that the mode shape errors may be a result of this.

The sensitivity of mode shapes to instrumentation errors is well documented in literature. Ewins (2000) highlights that accurate mode shape extraction demands a high degree of measurement precision across all DOFs, as any systematic error in the FRF amplitude or phase directly corrupts the identified eigenvectors. This is further supported by (Civera and Surace, 2024) that finds **that instrumentation related errors disproportionately affect mode shapes compared to natural frequencies**, this is consistent with the discrepancies that we observed in the current work.

Associated Code

All relevant codes are submitted together with this file in google classroom.
The programming languages used are Python and MATLAB.
Libraries used: scipy (scipy.linalg, scipy.signal), numpy, pandas, matplotlib.

References

Civera, M. and Surace, C. (2024). On the effect of intra- and inter-node sampling variability on operational modal parameters in a digital mems-based accelerometer sensor network for SHM: A preliminary numerical investigation. *Sensors*, 24.

Ewins, D. (2000). *Modal Testing: Theory, Practice and Application*. Research Studies Press Ltd., Baldock, Hertfordshire, England, 2nd edition.